Spherical-Ballistic Motion in a Central Gravitational Field

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Abstract

Projectile motion on a spherical body is fundamentally different from the classical flat-Earth parabolic approximation. When gravity follows an inverse-square law, the trajectory becomes a Keplerian conic section. This paper develops the complete mathematical theory of spherical-ballistic motion between arbitrary spherical-coordinate locations. Exact expressions are derived for orbital geometry, times of ascent and descent, flight times, energy, angular momentum, and target interception. Applications are discussed in warfare, missile defense, economics, finance, political science, international relations, artificial intelligence, data science, and strategic planning.

The paper ends with "The End"

Contents

1 Introduction

Classical projectile motion assumes a constant gravitational field.

Such approximations become inaccurate when:

1. ranges become large,
2. altitudes become significant,
3. planetary-scale trajectories are considered,
4. ballistic missiles are analyzed,
5. orbital interception problems arise.

The proper physical model is

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}$$

where

$$\mu = GM.$$

This equation yields conic-section trajectories, forming the basis of orbital mechanics, strategic missile design, spaceflight, and long-range warfare economics.

2 Geometry

Let

$$S = (r_1, \theta_1, \phi_1)$$

and

$$D = (r_2, \theta_2, \phi_2)$$

be two points in spherical coordinates.

Their Cartesian representations are

$$\mathbf{S} = \begin{pmatrix} r_1 \sin\theta_1 \cos\phi_1 \\ r_1 \sin\theta_1 \sin\phi_1 \\ r_1 \cos\theta_1 \end{pmatrix}$$

and

$$\mathbf{D} = \begin{pmatrix} r_2 \sin\theta_2 \cos\phi_2 \\ r_2 \sin\theta_2 \sin\phi_2 \\ r_2 \cos\theta_2 \end{pmatrix}.$$

The central angle between them is

$$\Delta = \cos^{-1}\left(\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2\right).$$

Expanding,

$$\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 = \sin\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2) + \cos\theta_1 \cos\theta_2.$$

3 Launch Conditions

Suppose an object is launched from S with speed s at elevation angle α above the local horizontal.

Then

$$v_r(0) = s \sin\alpha$$

and

$$v_\psi(0) = s \cos\alpha.$$

The specific angular momentum is

$$h = r_1 s \cos\alpha.$$

4 Conservation Laws

4.1 Energy

The specific orbital energy is

$$E = \frac{s^2}{2} - \frac{\mu}{r_1}.$$

4.2 Angular Momentum

Angular momentum remains constant:

$$h = r^2 \dot{\psi}.$$

Theorem 1. *The trajectory of a spherical ballistic projectile under inverse-square gravity is a conic section.*

Proof. Combining

$$h = r^2 \dot{\psi}$$

with

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}$$

and introducing

$$u = \frac{1}{r}$$

yields Binet's equation

$$u'' + u = \frac{\mu}{h^2}.$$

The general solution is

$$u = \frac{\mu}{h^2}\left[1 + e\cos(\psi - \omega)\right].$$

Inverting gives

$$r(\psi) = \frac{h^2/\mu}{1 + e\cos(\psi - \omega)}.$$

This is the polar equation of a conic section. □

5 Orbital Geometry

Define

$$p = \frac{h^2}{\mu}.$$

The trajectory is

$$r(\psi) = \frac{p}{1 + e\cos(\psi - \omega)}.$$

The eccentricity satisfies

$$e = \sqrt{1 + \frac{2Eh^2}{\mu^2}}.$$

Substituting yields

$$e = \sqrt{1 + \frac{r_1^2 s^2 \cos^2\alpha}{\mu^2}\left(s^2 - \frac{2\mu}{r_1}\right)}.$$

6 Apoapsis and Periapsis

The minimum radius is

$$r_p = \frac{p}{1 + e}.$$

The maximum radius is

$$r_a = \frac{p}{1 - e}.$$

Maximum altitude becomes

$$H = r_a - r_1.$$

7 Time of Ascent

For elliptic trajectories,

$$a = -\frac{\mu}{2E}.$$

The eccentric anomaly satisfies

$$r = a(1 - e\cos E_c).$$

Kepler's equation is

$$M = E_c - e\sin E_c.$$

The time is

$$t = \sqrt{\frac{a^3}{\mu}} M.$$

At apoapsis,

$$E_c = \pi.$$

Therefore

$$t_a = \sqrt{\frac{a^3}{\mu}} \left[\pi - (E_{c0} - e\sin E_{c0})\right].$$

8 Time of Descent

For symmetric launch and return radii,

$$t_d = t_a.$$

Hence

$$T = 2t_a.$$

9 Target Interception

The target condition is

$$r_2 = \frac{p}{1 + e\cos(\Delta - \omega)}.$$

Flight time is

$$T = \sqrt{\frac{a^3}{\mu}}(M_D - M_0).$$

10 Equation of Motion

The trajectory equation is

$$r(\psi) = \frac{h^2/\mu}{1 + e\cos(\psi - \omega)}.$$

The temporal relation is

$$r^2 \frac{d\psi}{dt} = h.$$

Integrating,

$$t(\psi) = \frac{1}{h} \int_0^\psi r(\varphi)^2 d\varphi.$$

Substitution yields

$$t(\psi) = \frac{p^2}{h} \int_0^\psi \frac{d\varphi}{[1 + e\cos(\varphi - \omega)]^2}.$$

11 Algorithmic Implementation

11.1 Pseudo-Code

```
Input:
r1, theta1, phi1
r2, theta2, phi2
s, alpha, mu

Compute:
S, D
Delta
h
E
e
p

Solve:
r2 = p/(1+e*cos(Delta-omega))

Compute:
a
E0
ED
M0
MD

Output:
trajectory
ta
td
T
```

12 Artificial Intelligence Applications

AI systems may use spherical-ballistic models for:

1. missile interception,
2. orbital targeting,
3. satellite collision avoidance,
4. autonomous defense systems,
5. reinforcement-learning navigation.

State vector:

$$x_t = (r, \psi, \dot{r}, \dot{\psi}).$$

Control vector:

$$u_t = (\Delta v_r, \Delta v_\psi).$$

Policy:

$$u_t = \pi(x_t).$$

13 Statistical Considerations

Launch uncertainty may be modeled as

$$s \sim N(\mu_s, \sigma_s^2)$$

and

$$\alpha \sim N(\mu_\alpha, \sigma_\alpha^2).$$

Impact probability becomes

$$P(\text{hit}) = \int I(\text{hit}) f(s, \alpha)\, ds\, d\alpha.$$

14 Economic Interpretation

Long-range ballistic capability changes strategic incentives.
Let

$$C_B$$

denote missile cost,

$$C_D$$

defense cost.
Expected strategic payoff:

$$U = P_H B - C_B - C_D.$$

Missile range depends directly upon the spherical-ballistic equations developed above.

15 Finance

Strategic defense expenditures influence sovereign risk.

Let

$$Y$$

be sovereign bond yield.

Then

$$Y = Y_0 + \lambda R$$

where

$$R$$

is geopolitical risk.

Ballistic capability influences

$$R.$$

16 Political Science

Ballistic reach changes:

1. deterrence structures,
2. alliance formation,
3. bargaining power,
4. coercive diplomacy.

17 International Relations

The ability to project force over spherical distances affects:

1. strategic stability,
2. escalation dynamics,
3. second-strike capability,
4. arms races.

18 Warfare Economics

The spherical-ballistic problem underlies:

1. artillery economics,
2. missile economics,
3. defense budgeting,
4. force projection.

19 Computational Complexity

Numerical solution typically requires:

1. root-finding,
2. Kepler equation inversion,
3. orbital propagation.

Typical complexity:

$$O(N)$$

for trajectory propagation.

20 Illustrative Table

Quantity	Symbol	Formula	Meaning
Energy	E	$\frac{s^2}{2} - \frac{\mu}{r_1}$	Specific energy
Angular Momentum	h	$r_1 s \cos\alpha$	Specific angular momentum
Eccentricity	e	$\sqrt{1 + \frac{2Eh^2}{\mu^2}}$	Orbit shape
Semi-major Axis	a	$-\frac{\mu}{2E}$	Orbit size
Flight Time	T	$\sqrt{\frac{a^3}{\mu}}(M_D - M_0)$	Travel duration

Table 1: Key quantities.

21 Conclusion

Spherical-ballistic motion under inverse-square gravity is governed by Keplerian conics. The exact theory provides closed-form expressions for trajectory geometry, energy, angular momentum, flight time, and interception. Applications extend beyond physics into economics, finance, political science, warfare economics, international relations, and artificial intelligence.

References

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Glossary

μ Gravitational parameter.

E Specific orbital energy.

h Specific angular momentum.

e Eccentricity.

a Semi-major axis.

p Semi-latus rectum.

r_p Periapsis radius.

r_a Apoapsis radius.

M Mean anomaly.

E_c Eccentric anomaly.

T Flight time.

The End